

1 Game Search

This is squarely in the ‘other solvers’ category, but I’ve had a few requests.

What do we mean by game search?

(tic-tac-toe example)

Minimax is the basic game search. The name comes from finding the *min* of the *max*.

(courtesy of Wikipedia, some pseudocode)

```
function minimax( node, depth, maximizingPlayer ) is
    if depth = 0 or node is a terminal node then
        return the heuristic value of node
    if maximizingPlayer then
        value := -infinity
        for each child of node do
            value := max( value, minimax( child, depth - 1, FALSE ) )
        return value
    else (* minimizing player *)
        value := +infinity
        for each child of node do
            value := min( value, minimax( child, depth - 1, TRUE ) )
        return value
(* Initial call *)
minimax( origin, depth, TRUE )
```

(also from Wikipedia, an example tree for a 2-player 2-option game with example leaf values)

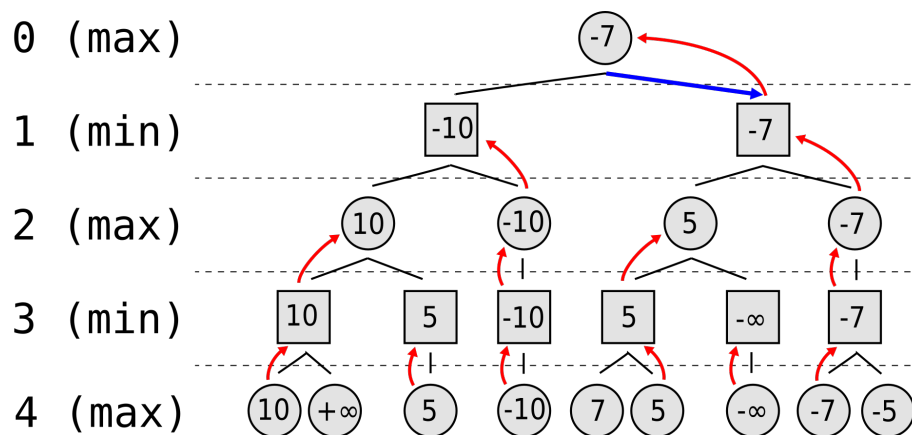


Figure 1: An example of a minimax tree for a two-player game. (By Nuno Nogueira (Nmnogueira) – own work using <http://en.wikipedia.org/wiki/Image:Minimax.svg>, created in Inkscape by author, CC BY-SA 2.5, <https://commons.wikimedia.org/w/index.php?curid=2276653>.)

Let's construct this tree with values for our tic-tac-toe example.

What does this tell us about games? How can we interpret the value at the root?

1.1 How can we make this faster?

Without improvements, minimax will construct the entire game tree.

We can deploy many tricks!

- alpha-beta pruning
- hash tables for repeat states
- node ordering (maybe find a win faster)
- trickery to make at-node computation faster
- if we're not likely to solve exactly, we can use node value heuristics

Two now-old things that were exciting at the time:

- Alpha-Go <https://en.wikipedia.org/wiki/AlphaGo>, via Monte Carlo Tree Search
- Monte Carlo Tree Search, more generally

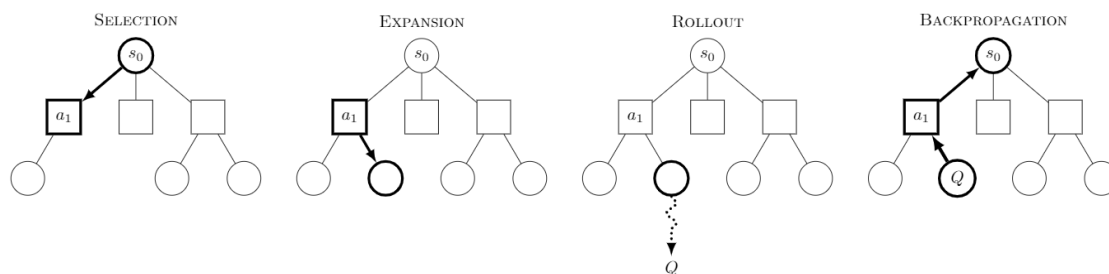


Figure 2: An example showing Monte Carlo Tree Search. (By Robert Moss – own work, CC BY-SA 4.0, <https://commons.wikimedia.org/w/index.php?curid=111182752>.)